

Test Before You Sleep: A Policy-Reactive Digital Twin for RAN Carrier Switch-Off

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Abstract

Energy consumption is a dominant operational cost and a chief sustainability concern for Mobile Network Operators (MNOs), and the Radio Access Networks (RANs) alone account for around 73% of total network energy expenditures. Switching off capacity Radio Frequency (RF) carriers at times of low service demand is the most immediate means to reduce the energy footprint of RAN infrastructure. However, every switch-off decision also removes wireless capacity and risks degrading Quality of Service (QoS). The possibility of upsetting customers precludes operators from trialing aggressive carrier control policies on live networks, making it arduous to adopt innovative RAN energy efficiency policies, such as those based on Machine Learning (ML). In this paper, we present a policy-reactive RAN Digital Twin (DT) that reproduces at scale realistic traffic demands, post-switch-off handovers, energy costs, and QoS performance. The DT lets an external policy submit switch-on/off actions, redistributes load across the RAN according to such actions, and outputs energy consumption and potential QoS degradation over an open interface. We validate each component of the proposed DT against operator measurements and the literature, instantiate the twin on a real city-scale deployment with 3,054 carriers, and expose the energy-QoS trade-offs that characterize baseline threshold policies.

CCS Concepts

• **Computing methodologies** → **Interactive simulation**; **Neural networks**; *Simulation evaluation*; • **Networks** → **Mobile networks**.

Keywords

digital twin, generative adversarial networks, GAN, synthetic data generation, mobile networks, network simulation

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1 Introduction

Reducing energy consumption is a primary goal for the mobile industry, with strong environmental and economic implications. Energy already accounts for between 20% and 40% of a mobile operator's operational expenditure [1], and the RAN alone is responsible for around 73% of the total energy an operator consumes [2].

Modern RANs are over-provisioned by design: several carriers on different bands cover the same area, with always-on coverage carriers on the low RF bands and switchable capacity carriers on the high bands [3, 4]. This makes switch-off possible but its effect hard to anticipate: the load of a carrier that sleeps does not disappear; it transfers via User Equipment (UE) handovers to the carriers that stay on, raising their utilization and potentially lowering the throughput their users perceive [4]. Quantifying this transfer, *i.e.*, the energy saved relative to any QoS degradation, is the core difficulty a switch-on/off policy faces.

Operators deal with such trade-offs today in a heterogeneous mix of technologies: some regions still rely mostly on 4G, while 5G is being deployed and commercial 6G is contemplated for 2030 [5]. Newer generations of cellular technology are more efficient per bit but also denser, so absolute energy can grow without active management [6], and carrier switch-on/off remains a common lever on the RAN side in production. Operators today largely rely on conservative threshold-based policies that sleep a few carriers overnight; recent production trials show these hit a clear barrier with the imposed constraint that no user may lose throughput [7], motivating more flexible policies and a safe way to test them.

Safely testing policies requires an environment that is realistic, runs at the scale of a real deployment, and responds to policy actions. Digital Twins (DTs) are appealing in this setting because they let operators anticipate load, test new configurations, and estimate their impact before changing

the production network. In this work, we focus on the configuration decision that matters most for RAN energy, *i.e.*, which carriers to switch on or off and when.

Existing tools each cover part of this need. High-fidelity emulators [8, 9] and channel simulators reproduce a single cell or a handful of cells at the protocol or signal level, but not a city. Operator-grade network twins [10, 11] reach city scale, but are built to mirror and forecast the network behavior under RAN updates rather than to take a switch-on/off policy as input and explore its consequences with respect to energy and QoS. Our work sits in a gap left by existing tools, *i.e.*, a city-scale and *policy-reactive* twin, focusing on the predictive, what-if side of network management. Our DT allows researchers and operators to leverage two main use cases: (i) simulate and thoroughly test the impact of an energy-saving policy and the tradeoff between reducing energy consumption and maintaining QoS before deploying it, and (ii) develop tailor-made new policies using ML techniques such as Reinforcement Learning. We make this positioning concrete in Section 2 (Table 1).

Two design choices make our solution versatile. (i) We derive load transfer from ray-tracing over public data, allowing the twin to run on arbitrary deployments without requiring an expensive measurement campaign. This contrasts with measurement-driven handover models that need per-user signal reports from the live network [4]. (ii) We emulate the RAN side only, *i.e.*, antenna-level Key Performance Indicators (KPIs) and capacity-carrier switch-on/off, leaving aside the core, the transport, and individual users. The twin is technology-agnostic, handling 4G and 5G NR carriers alike. **Contributions.** We make three contributions. (i) We build a RAN DT in which realistic demands are produced by a context-conditioned generative model, complemented by measurement-grounded handover and energy/QoS models. (ii) We close a policy-reactive loop, *i.e.*, demand, action, load transfer, and energy/QoS, behind a REST/OpenAPI interface that serves as the interaction point for externally trained or reinforcement-learning energy-saving agents. (iii) We instantiate the twin on a real city-scale deployment of 3,054 carriers in Brussels, validate each model against operator measurements and the literature, and characterize the energy-QoS trade-offs of threshold policies. The twin supports both 4G and 5G NR and accepts any user-supplied deployment; we offer a ready-to-run 4G reference deployment, the most detailed configuration our data-sharing agreements allow us to release publicly. The rest of the paper presents the twin’s architecture (Section 2), its three models and their validation (Section 3), and the city-scale evaluation (Section 4).

Table 1: Our DT versus tools for RAN experimentation.

Platform	City scale	Real demand	Load transfer	Energy & QoS	Policy loop	Open API	PHY fidelity
Network simulators (ns-3) [8]	✓						
Protocol emulators/testbeds [9]					✓		✓
Near-RT RIC (O-RAN) [12]					✓	✓	
Physics RF twins [10, 11]	✓						✓
Operator planning twins [13]	✓					✓	
Our RAN DT	✓	✓	✓	✓	✓	✓	

2 The Digital Twin

We offer a comparison with existing tools and platforms, highlighting the different dimensions targeted, followed by an overview of the architecture.

2.1 Positioning

Network simulators such as ns-3 scale to large topologies, but they model abstract protocols rather than commercial RANs and carry no notion of operator demand, energy, or QoS [8]. Protocol emulators and testbeds, from software stacks to SDR platforms such as Colosseum, reproduce the radio stack with high fidelity, but are limited to the scale of a cell or a small testbed [9]. Near-real-time controllers in the O-RAN architecture react to the network within sub-second loops, but they operate at campus scale and target scheduling rather than energy-saving carrier switch-off [12].

Two recent families come closer to our setting. Physics-accurate radio twins, *e.g.*, NVIDIA Aerial Omniverse and Nokia’s RAN DT, simulate propagation from a single tower up to city scale for 6G research, but they reproduce the *full* physical layer rather than the operator-level interplay of demand, handovers, energy, and QoS [10, 11]. Operator automation and planning platforms, such as Ericsson’s Intelligent Automation Platform (EIAP), run at network scale and expose open interfaces for control applications, but they act on the live network rather than emulating the consequences of a policy in a safe sandbox [13]. In contrast to these, our twin drives realistic demands through a measurement-grounded handover model at city scale and produces energy and QoS metrics for an externally supplied policy. *We trade the protocol-level fidelity of emulators for operational scale and a closed policy loop.*

2.2 Architecture

Figure 1 illustrates how the twin operates. It is initialized with a description of the target RAN deployment (①): the location of each site, the carriers deployed at each site, and

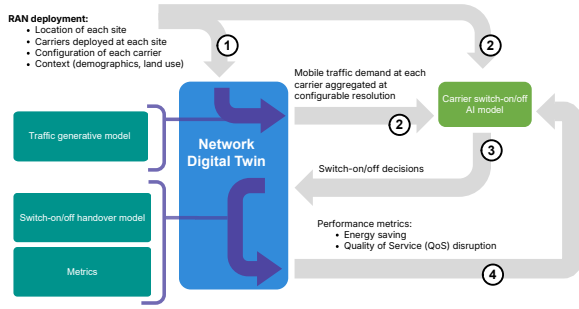


Figure 1: The Digital Twin, its components, and its closed loop with an external switch-on/off policy.

Table 2: The Brussels reference deployment.

Topology	
Area (deployment footprint)	~140 km ²
3D scene elements (walls, roofs, ...)	~556,000
Sites / sectors	288 / 862
Spectrum (4G)	
800 MHz / 1800 MHz (coverage)	1,041 / 859
2100 MHz / 2600 MHz (capacity)	604 / 550
Channel bandwidths	10, 15, 20 MHz
Context	
Grid per site (5×5, 0.25 km ²)	25 cells
Features per cell	8
Total context grid cells	7,200

the configuration of every carrier, *i.e.*, frequency band, bandwidth, transmit power, azimuth, and height. The traffic-generating model also needs context about the area each carrier serves, *e.g.*, population density, land use, and building density. For deployments in Western Europe, we fetch this context automatically from open repositories such as WorldPop, Copernicus Urban Atlas, and OpenStreetMap; elsewhere, the context can be provided alongside the deployment information. The twin supports deployments containing an arbitrary mix of 4G and 5G NR carriers. To facilitate immediate exploration of the twin’s capabilities, we ship our RAN DT with a ready-to-run reference 4G deployment of 3,054 carriers in Brussels, of which 1,153 are capacity carriers, together with all the information its models need (Table 2).

Once initialized, the twin runs a closed loop with an external switch-on/off policy. At each step, it exposes the mobile traffic demand at every carrier, aggregated at a configurable temporal resolution (②), which the policy would consume to produce decisions about the carriers to switch off or on under that demand (③). The twin then applies the decisions: it redistributes the load of each switched-off carrier across the carriers that stay on through the handover model, recomputes the network state, and reports the performance metrics,

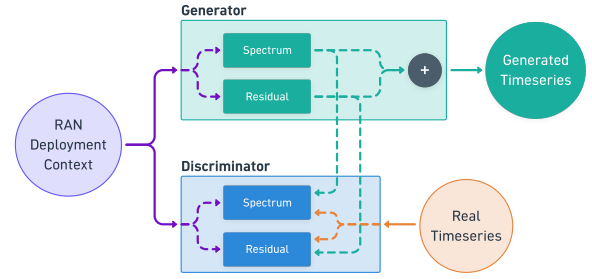


Figure 2: Context-conditioned GAN: a spectrum and a residual branch generate the per-carrier series, which a matching discriminator scores against real traffic.

i.e., the energy saved and the QoS degradation relative to an all-on baseline (④). We describe in Section 3 the three models inside our RAN DT, namely: the traffic generation, the handover, and the energy and QoS models.

The loop is technology-agnostic: a carrier may be 4G or 5G NR, and switch-on/off decisions act on capacity carriers, while coverage carriers stay on to preserve connectivity.

Open interface. The policy and the twin communicate over a REST/OpenAPI interface: the policy submits the switch-on/off actions and reads the network state, the demand, and the metrics through GET endpoints. This separation is what turns the twin into a *playground*: we supply the environment and the measurement, and any controller, from the threshold rule we use in Section 4 to a reinforcement-learning agent [14], interacts with the digital twin via the same interface. Each simulation is independent; therefore, several policies can be trained and evaluated in parallel.

3 Models and Their Validation

The twin relies on three measurement-grounded models described next, along with details of how each is validated.

3.1 Realistic Demand Generation

The first model generates the traffic demand that drives the twin. For each carrier, we require realistic time series of served traffic, even when no ground truth measurements exist for it. These series are synthesized with a generative adversarial network (GAN) conditioned on the context of each carrier: the territory it serves, encoded from population density, land use, and road presence, together with antenna metadata such as frequency band and azimuth. The model targets the volume of traffic served by each carrier, *i.e.*, the antenna-level KPI on which switch-on/off decisions are taken; it does not reproduce individual users, which would add detail the energy use case does not need.

Our generator is primarily built on the notion of spectral decomposition (Figure 2). Inspired by SpectraGAN [15], we generate each series in the frequency domain rather than

sample-by-sample, thereby capturing the strong daily periodicity of mobile traffic. Critically, unlike SpectraGAN, we synthesize realistic traffic demands for each carrier individually and independently of the rest of the target RAN, allowing us to seamlessly switch off specific carriers without affecting the steady traffic generation process at all other carriers.

To that end, a context encoder, consisting of a Multilayer Perceptron (MLP) with a feature-wise linear modulation (FiLM) layer, maps per-carrier contexts into hidden representations that condition our two generator components: (i) a *spectrum generator* that produces the dominant frequency components of the series, mapped back to the time domain through an inverse FFT, and (ii) a *residual generator*, a recurrent network, that adds the finer fluctuations. The synthetic series is the sum of the two. To keep adversarial training stable, we apply spectral normalization on the discriminator and a two-time-scale update rule. A statistical profiling module further clusters carriers by context and feeds the empirical traffic distribution of each cluster back into the model, both as an additional conditioning input and as a loss term, so that the generated series reflects the variability observed within structurally similar carriers.

We validate the generator with k -fold cross-validation across carriers, comparing synthetic and real KPIs along two axes. First, distribution similarity: for each carrier, we measure how much the synthetic and real value distributions overlap, using histogram intersection and the histogram Earth Mover’s Distance (EMD). Second, time-aligned error: we report the mean absolute and mean squared error between the synthetic and ground truth series. We repeat this across an ablation of the conditioning context, *i.e.*, spatial features only, spatial features plus antenna metadata, and both plus statistical profiles. Distribution overlap improves, and the error decreases as the context grows richer: the median EMD drops from 6.41 with spatial features alone to 3.53 once antenna metadata and statistical profiles are added, as shown in Figure 3, top. For the final model that uses the full feature set, the generated series and their value distributions closely track the ground truth, as shown for two example carriers in Figure 3 (bottom).

3.2 Handover-Driven Load Transfer

When a capacity carrier is switched off, its users do not vanish; they are handed over to nearby carriers, and where they land is far from uniform. Measurements of switch-off events in a production network reveal a heterogeneous pattern: most users transfer to a carrier in the same sector, a smaller share to another sector of the same site, and the rest to nearby sites, with the probability decaying with distance [7]. Capturing this transfer matters for two reasons. First, the load that moves changes the energy the carriers left on must draw, so

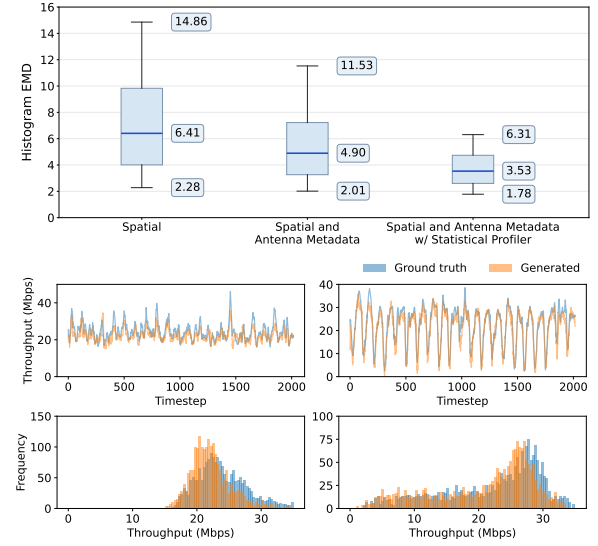


Figure 3: Generator validation. Top: histogram EMD (lower is better) between synthetic and real KPIs across three context sets. Bottom: using the final model with all the features, two example carriers with their generated and ground truth series and value distribution.

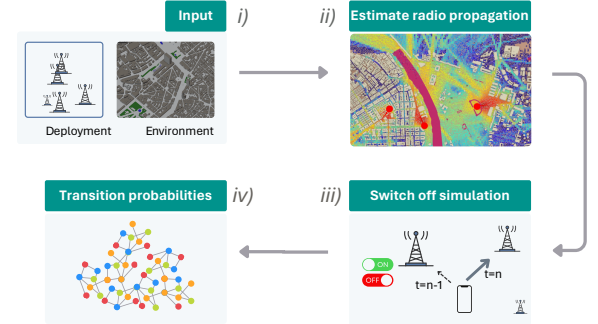


Figure 4: Main steps of the handover model.

ignoring it biases the energy estimate [4]. Second, and more importantly, it drives QoS: a switch-off can look attractive for energy alone while overloading a neighboring carrier, trading a large QoS penalty for a small saving [7].

Previous approaches [4, 16] tend to sidestep this complexity: either they assume each switched-off carrier hands off its users to a single predetermined carrier in the same sector *that can cope with the new load*, or they avoid the transfer altogether and reason about energy in isolation [4]. Data-driven models that capture it accurately rely on per-user signal measurement reports collected from the live network [4], which an operator can gather at nontrivial cost, but a twin meant to run on an arbitrary deployment cannot assume. Instead, we estimate the transfer using a physics-based propagation model on public data; therefore, the twin needs no measurement campaign.

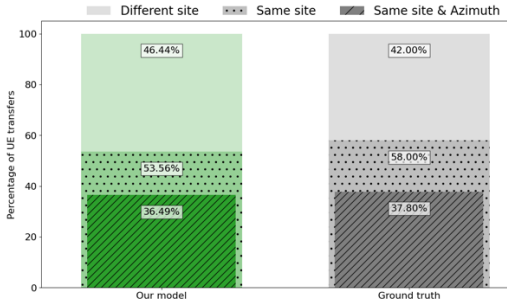


Figure 5: Where users land after a switch-off: the modeled transfer breakdown across same sector (site & azimuth), same site, and different site, against operator measurements.

Figure 4 shows the four steps. *i)* We start from the RAN deployment and the surrounding environment, *i.e.*, the 3D buildings, materials, parks, and water bodies that shape propagation, fetched from open data as in Section 2. *ii)* We feed both to the Sionna-RT ray-tracing model [17], which estimates per-antenna signal metrics such as received signal strength, path gain, and Signal-to-Interference-Noise Ratio (SINR) across the area. *iii)* We then simulate a switch-off: a user attached to the target antenna is re-associated with the strongest remaining antenna after the target antenna is removed. *iv)* Repeating this over many locations and antennas yields a directed graph whose edges have an associated value $P(j_{t=n} | i_{t=n-1})$, which is the probability that a user on antenna i transfers to antenna j when i switches off. At runtime, the twin redistributes the load of each switched-off carrier among the active antennas according to these probabilities. We compare the resulting transfer breakdown against operator switch-off measurements (Figure 5), and the model reproduces where users land: 53.6% of transfers stay within the same site (measured 58%), of which 36.5% reach a carrier in the same sector and azimuth (measured 37.8%), while 46.4% cross to a different site (measured 42%). This agreement, obtained without any per-user measurement campaign, enables the twin to redistribute load credibly, account for potential impacts on QoS, and produce accurate energy consumption estimates.

3.3 Energy and QoS Modeling

The twin scores each network state with two metrics.

Energy. We estimate energy with the measurement-based base station power model of [18], calibrated on city-scale 4G and 5G data. The energy of a base station splits into two,

$$E_{BS} = E_{BBU} + \sum_i E_{RRU}^i, \quad (1)$$

where E_{BBU} is the energy spent processing baseband signals, which stays roughly constant and independent of load, and

E_{RRU}^i is the energy of the radio unit serving carrier i ,

$$E_{RRU} = \alpha \cdot R_{PRB} \cdot P_m + \gamma, \quad (2)$$

with R_{PRB} the fraction of resource blocks in use (the load), P_m the maximum transmit power set by the operator, and α and γ the power-amplifier efficiency and fixed circuit power, specific to the carrier's generation, frequency, and number of transmitters. Coverage carriers stay on, so E_{BBU} is unaffected by switch-off; the savings come entirely from the RRU term, which drops to zero ($R_{PRB} = 0$, $\gamma = 0$) when a capacity carrier sleeps. Figure 6 shows the resulting relationship between a carrier's daily downtime and its energy: energy decreases with downtime, and its variance grows as carriers stay on longer, reflecting the heterogeneity of the daily traffic loads they then carry. As with every component, the parameters can be re-fit to an operator's own measurements without touching the rest of the twin.

QoS. The twin needs a per-carrier mapping from the KPIs it produces, *e.g.*, the carrier load, to a quality signal that exposes the cost of a switch-off. There is no consensus on how to quantify QoS from network-side data: the right definition is operator-dependent and tied to the services (*e.g.*, video, voice, browsing) and to the thresholds each operator optimizes for and cares about. The production metrics operators use combine utilization, throughput, and failure KPIs into a synthetic *hotspot* score that ranks underperforming sectors, with the exact KPI set, weights, and thresholds tuned empirically from vendor and operator domain knowledge and drifting as services evolve [19].

What is consistent is that high resource utilization is treated as the onset of congestion and degraded service, typically once Physical Resource Block (PRB) utilization reaches the 75–80% range [7, 19]. Rather than hardcode one operator's approach, the twin offers a flexible QoS metric grounded on two ideas. First, per-user throughput degrades non-linearly with load and collapses as a carrier approaches saturation [20, 21], meaning that quality stays flat at low load and falls sharply past a critical point. Second, this soft-QoS relationship between resource availability and quality is well captured by a sigmoid with an inflection at that critical point [22], which matches measured experience better than a binary threshold [19].

Concretely, we map a carrier's load x , *i.e.*, its PRB utilization in percentages, to a score on a 1 to 5 scale (the higher the better) through an inverted logistic model,

$$q(x) = L + \frac{U - L}{1 + e^{k(x - x_0)}}, \quad (3)$$

with floor $L = 1$, ceiling $U = 5$, critical load x_0 (default 75%), and decay rate k (default 0.22): QoS stays near U at low load and decays toward L once load passes x_0 .

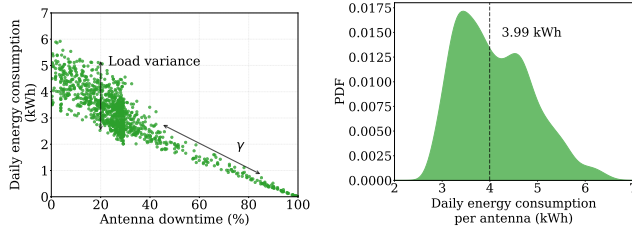


Figure 6: Radio-unit energy (E_{RRU}) against carrier downtime: capacity carriers (left) and always-on coverage carriers (right).

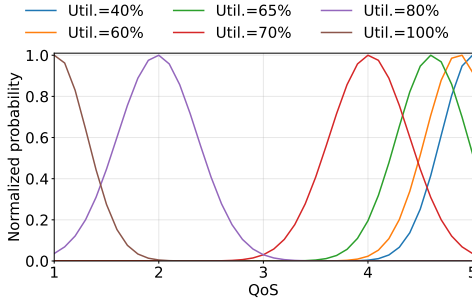


Figure 7: Sampled QoS as a function of antenna load based on an inverted logistic model.

To capture the variability of a real switch-off, we do not use $q(x)$ directly but sample the score from a normal distribution centered on $q(x)$, with a standard deviation that scales with the local slope; the spread is largest around the critical load, where small load changes swing QoS the most, and smallest at the extremes. Figure 7 shows the resulting QoS distribution as a function of load. This model is a pluggable component: users can supply their own metric or tune x_0 , k , and the spread parameters to match their requirements.

4 City-Scale Evaluation

We instantiate the twin on a scenario of a major operator's deployment in the Brussels-Capital region: 3,054 carriers across 288 sites and 862 sectors over roughly 140 km² (Table 2). The lower bands (800 and 1800 MHz) provide coverage and stay on, while the higher bands (2100 and 2600 MHz) are the 1,153 capacity carriers eligible for switch-off. We build the scenario from public RAN data and the per-site territory context the generator needs, a 5×5 grid of population, land-use, and road features. We then run a day at 15-minute resolution under an all-on baseline and three fixed-threshold policies: a policy switches a sector's capacity carriers off once the sector's average load falls below an off-threshold and back on above an on-threshold, and we test 30/60 and 50/80 all day, plus 50/80 restricted to 20:00–06:00.

Figure 8 shows a day of these policies. The energy panel (left) shows that a more aggressive switch-off policy saves more energy: 50/80 cuts aggregate energy the most during

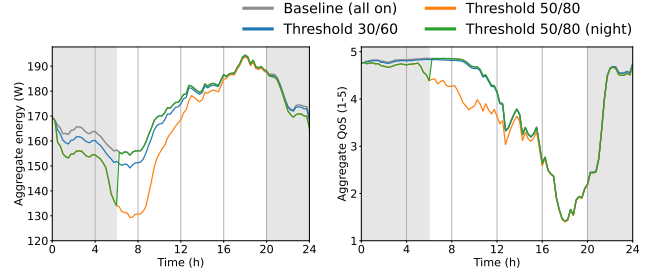


Figure 8: A day of the twin on the Brussels deployment under three threshold policies and the all-on baseline: aggregate energy (left) and QoS (right). Shaded bands mark the night window (20:00–06:00).

the low-demand night and early morning, while the night-only policy saves only inside its window and tracks the baseline by day. The QoS panel (right) shows the cost: 50/80 is the only policy that visibly lowers QoS, and only while it switches carriers off under rising load (the morning ramp), whereas the gentler 30/60 and the night-only policy stay on top of the baseline; around the midday peak, all policies converge, as the high load saturates the network regardless of the policy. Together, the panels make the trade-off concrete: over the day, 50/80 saves 13% of the energy per switched-off carrier at a QoS cost (decreasing from 3.9 to 3.7 on the 1 to 5 scale), and a fixed threshold cannot reach those savings without that cost, which is exactly the trade-off a learned policy could navigate on this twin.

5 Conclusion

We presented a policy-reactive RAN DT that enables operators and researchers to test carrier switch-on/off policies on real, city-scale deployments and observe their energy and QoS consequences before rollout in production networks. The twin closes a loop that existing tools leave open: it pairs realistic demand with a ray-traced handover model (measurement-grounded), physics-based energy and QoS metrics, and exposes them through an open interface where any controller, from a threshold-based rule system to a learned agent, can attach. In the Brussels deployment we exemplify, it makes the energy-QoS trade-off concrete, showing that a fixed-threshold, aggressive policy has an associated quality cost.

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References

- [1] A. De Domenico *et al.*, “Advancing green mobile networks,” *nature communications*, 2026.
- [2] GSMA Intelligence, “Going green: Benchmarking the energy efficiency of mobile,” *London, UK: GSMA*, 2021.
- [3] D. Lopez-Perez *et al.*, “A survey on 5g radio access network energy efficiency: Massive mimo, lean carrier design, sleep modes, and machine learning,” *IEEE Communications Surveys and Tutorials*, vol. 24, pp. 653–697, 2022.
- [4] A. De Domenico *et al.*, “Modeling user transfer during dynamic carrier shutdown in green 5g networks,” *IEEE Transactions on Wireless Communications*, vol. 22, pp. 5536–5549, 8 2023.
- [5] Ericsson, “Ericsson Mobility Report,” Ericsson, Tech. Rep., November 2025.
- [6] GSMA, “5G Energy Efficiencies: Green is the New Black,” GSMA, White Paper, 2020.
- [7] O. E. Martínez-Durive *et al.*, “An evaluation of ran sustainability strategies in production networks,” in *IEEE INFOCOM*, 2025.
- [8] T. R. Henderson *et al.*, “Network simulations with the ns-3 simulator,” in *SIGCOMM Demonstration*, 2008.
- [9] D. Villa *et al.*, “Colosseum as a digital twin: Bridging real-world experimentation and wireless network emulation,” *IEEE TMC*, vol. 23, no. 10, 2024.
- [10] Nokia, “Nokia RAN Digital Twin, powered by NVIDIA Aerial Omniverse Digital Twin,” 2025.
- [11] NVIDIA, “Aerial Omniverse Digital Twin,” 2024.
- [12] A. Garcia-Saavedra *et al.*, “O-ran: Disrupting the virtualized ran ecosystem,” *IEEE Communications Standards Magazine*, vol. 5, no. 4, 2021.
- [13] Ericsson, “Ericsson Intelligent Automation Platform (EIAP),” 2024, available: <https://www.ericsson.com/en/ran/intelligent-ran-automation/intelligent-automation-platform>.
- [14] J. S. Pujol-Roigl *et al.*, “Deep reinforcement learning for cell on/off energy saving on wireless networks,” in *2021 IEEE GLOBECOM*, 2021.
- [15] K. Xu *et al.*, “Spectragan: Spectrum based generation of city scale spatiotemporal mobile network traffic data,” in *ACM CoNext*, 2021.
- [16] M. Ajmone Marsan *et al.*, “Optimal energy savings in cellular access networks,” in *2009 IEEE INFOCOM Workshops*, 2009, pp. 1–5.
- [17] J. Hoydis *et al.*, “Sionna: An open-source library for next-generation physical layer research,” arXiv:2203.11854, 2022.
- [18] Y. Ma *et al.*, “Sustainable connections: Exploring energy efficiency in 5g networks,” in *ACM CoNEXT*, 2024.
- [19] I. Leontiadis *et al.*, “The good, the bad, and the kpis: How to combine performance metrics to better capture underperforming sectors in mobile networks,” in *IEEE ICDE*, 2017.
- [20] T. Bonald and A. Proutiere, “Wireless downlink data channels: User performance and cell dimensioning,” in *ACM MobiCom*, 2003.
- [21] J. W. Roberts, “A survey on statistical bandwidth sharing,” *Computer networks*, vol. 45, no. 3, pp. 319–332, 2004.
- [22] G. Lee and other, “Qoe-aware scheduling for sigmoid optimization in wireless networks,” *IEEE Communications Letters*, vol. 18, no. 11, 2014.